

# Practice Task 1

## Demo File for Data Preparation and Cleaning.

Description: We were tasked to use Excel and practice every function and command for us to proceed in another task. This task introduces Data Cleaning, Different Data such as Star Schema, Snowflake Schema.

Screenshots of the tasks:

## 1.Excel Practice

[illegible]

2.Data to be cleaned

| Date      | Client                                   | Contact         | Department          | Payment  | Revenue | Profit  | Profit Margin |
|-----------|------------------------------------------|-----------------|---------------------|----------|---------|---------|---------------|
| 5/30/2023 | AMAZON.COM, INC. (XNAS:AMZN)             | Bill Smith      | Cloud Tech_Texas    | Transfer | \$4,500 | \$598   | 13.3%         |
| 5/30/2023 | TESLA, INC. (XNAS:TSLA)                  | KEN Singh       | Strategy_New York   | PayPal   | \$3,800 | \$1,045 | 27.5%         |
| 5/30/2023 | NETFLIX, INC. (XNAS:NFLX)                | Harley Fritz    | Strategy_New York   |          | \$3,713 | \$1,009 | 27.2%         |
| 5/30/2023 | THE PROCTER & GAMBLE COMPANY (XNYS:PG)   | Nyla Novak      | Operations_Florida  |          |         | \$779   | #DIV/0!       |
| 5/30/2023 | THE GOLDMAN SACHS GROUP, INC. (XNYS:GS)  | David Rasmussen | Operations_Florida  | Check    | .       | \$684   | #VALUE!       |
| 5/31/2023 | JPMORGAN CHASE & CO. (XNYS:JPM)          | IVAN HINEY      | Cloud Tech_Texas    | Transfer | \$6,100 | \$544   | 8.9%          |
| 5/31/2023 | MORGAN STANLEY (XNYS:MS)                 | JONha Ma        | Cloud Tech_Texas    | Transfer | \$4,625 | \$670   | 14.5%         |
| 5/31/2023 | CITIGROUP INC. (XNYS:C)                  | Jordan Boone    | Cloud Tech_Texas    | Transfer | \$3,800 | \$2,045 | 53.8%         |
| 5/31/2023 | BANK OF AMERICA CORPORATION (XNYS:BAC)   | Kylee Townsend  | Cloud Tech_Texas    | Card     | \$3,600 | \$1,564 | 43.4%         |
| 5/31/2023 | WALMART INC. (XNYS:WMT)                  | Nora Rollins    | Cloud Tech_Texas    | Check    | \$5,100 | \$1,220 | 23.9%         |
| 5/31/2023 | TARGET CORPORATION (XNYS:TGT)            | BRENDAN Wallace | Cloud Tech_Texas    | Check    | \$4,750 | \$1,435 | 30.2%         |
| 5/31/2023 | COSTCO WHOLESALE CORPORATION (XNAS:COST) | Conor Wise      | Operations_Florida  | Transfer | \$6,000 | \$998   | 16.6%         |
| 5/31/2023 | MCDONALD'S CORPORATION (XNYS:MCD)        | Steven Michael  | Big Data_California | Check    | \$4,500 | \$780   | 17.3%         |
| 6/1/2023  | EXXON MOBIL CORPORATION (XNYS:XOM)       | Lucia Mckay     | Big Data_California | Card     |         | \$1,044 | #DIV/0!       |
| 6/1/2023  | VERIZON COMMUNICATIONS INC. (XNYS:VZ)    | JoSE Roach      | Big Data_California | Transfer | \$3,713 | \$1,222 | 32.9%         |
| 6/1/2023  | THE HOME DEPOT, INC. (XNYS:HD)           | Franklin WRIGT  | Big Data_California | Transfer | \$4,950 | \$1,065 | 21.5%         |
| 6/1/2023  | CISCO SYSTEMS, INC. (XNAS:CSCO)          | Alia Thornton   | Operations_Florida  | Transfer | \$4,750 | \$810   | 17.1%         |
| 6/1/2023  | CHEVRON CORPORATION (XNYS:CVX)           | Denzel Flores   | Operations_Florida  | Transfer | \$7,320 | \$933   | 12.7%         |
| 5/31/2023 | COSTCO WHOLESALE CORPORATION (XNAS:COST) | Conor Wise      | Operations_Florida  | Transfer | \$6,000 | \$998   | 16.6%         |
| 5/31/2023 | MCDONALD'S CORPORATION (XNYS:MCD)        | Steven Michael  | Big Data_California | Check    | \$4,500 | \$780   | 17.3%         |
| 6/1/2023  | AT&T INC. (XNYS:T)                       | Bruno CordOVA   | Big Data_California | Transfer | \$5,088 | \$655   | 12.9%         |
| 6/1/2023  | INTEL CORPORATION (XNAS:INTC)            | jaylynn napp    | Big Data_California | Transfer | \$4,500 | \$722   | 16.0%         |
| 6/1/2023  | GENERAL MOTORS COMPANY (XNYS:GM)         | Bruce RICH      | Big Data_California | Card     | \$4,250 | \$901   | 21.2%         |
| 6/2/2023  | MICROSOFT CORPORATION (XNAS:MSFT)        | Arturo Moore    | Big Data_California | PayPal   | \$5,250 | \$1,349 | 25.7%         |
| 6/2/2023  | COMCAST CORPORATION (XNAS:CMCSA)         | Bryce Carpenter | Strategy_New York   | PayPal   | \$6,500 | \$1,288 | 19.8%         |
| 6/2/2023  | DELL TECHNOLOGIES INC. (XNYS:DELL)       | Jaidyn ANDERSEN | Strategy_New York   | PayPal   | \$7,500 | \$1,664 | 22.2%         |
| 6/2/2023  | JOHNSON & JOHNSON (XNYS:JNJ)             | MARK WALM       | Strategy_New York   | Transfer | \$5,500 | \$1,320 | 24.0%         |
| 6/2/2023  | FEDEX CORPORATION (XNYS:FDX)             | HARRY LEE       | Strategy_New York   | Transfer | \$4,625 | \$1,001 | 21.6%         |
| 6/2/2023  | GENERAL ELECTRIC COMPANY (XNYS:GE)       | JOSH JOHNSON    | Strategy_New York   | Transfer | \$4,500 | \$960   | 21.3%         |
| 6/2/2023  | LOCKHEED MARTIN CORPORATION (XNYS:LMT)   | Mik Naam        | Strategy_New York   | Card     | \$5,400 | \$540   | 10.0%         |
| 5/30/2023 | THE GOLDMAN SACHS GROUP, INC. (XNYS:GS)  | David Rasmussen | Operations_Florida  | Check    | \$5,000 | \$684   | 13.7%         |

3. REVENUE\_FACT

| REVENUE AND PROFIT INFO |       | Date      | client_id | dept_id | state_id | payment_id | revenue    | profit     | margin |
|-------------------------|-------|-----------|-----------|---------|----------|------------|------------|------------|--------|
| 1                       | t-101 | 5/30/2023 | c-101     | d-101   | s-101    | p-101      | \$4,500.00 | \$598.00   | 13%    |
| 1                       | t-102 | 5/30/2023 | c-102     | d-102   | s-102    | p-102      | \$3,800.00 | \$1,045.00 | 28%    |
| 1                       | t-103 | 5/30/2023 | c-103     | d-102   | s-102    | p-103      | \$3,712.50 | \$1,009.00 | 27%    |
| 1                       | t-104 | 5/30/2023 | c-104     | d-102   | s-103    | p-103      | \$4,994.00 | \$779.00   | 16%    |
| 1                       | t-105 | 5/30/2023 | c-105     | d-103   | s-103    | p-104      | \$5,000.00 | \$684.00   | 14%    |
| 1                       | t-106 | 5/31/2023 | c-106     | d-101   | s-101    | p-101      | \$6,100.00 | \$544.00   | 9%     |
| 1                       | t-107 | 5/31/2023 | c-107     | d-101   | s-101    | p-101      | \$4,625.00 | \$670.00   | 14%    |
| 1                       | t-108 | 5/31/2023 | c-108     | d-101   | s-101    | p-101      | \$3,800.00 | \$2,045.00 | 54%    |
| 0                       | t-109 | 5/31/2023 | c-109     | d-101   | s-101    | p-105      | \$3,600.00 | \$1,564.00 | 43%    |
| 1                       | t-110 | 5/31/2023 | c-110     | d-101   | s-101    | p-104      | \$5,100.00 | \$1,220.00 | 24%    |
| 2                       | t-111 | 5/31/2023 | c-111     | d-101   | s-101    | p-104      | \$4,750.00 | \$1,435.00 | 30%    |
| 3                       | t-112 | 5/31/2023 | c-112     | d-103   | s-103    | p-101      | \$6,000.00 | \$998.00   | 17%    |
| 4                       | t-113 | 5/31/2023 | c-113     | d-104   | s-104    | p-104      | \$4,500.00 | \$780.00   | 17%    |
| 5                       | t-114 | 6/1/2023  | c-114     | d-104   | s-104    | p-105      | \$4,994.00 | \$1,044.00 | 21%    |
| 6                       | t-115 | 6/1/2023  | c-115     | d-104   | s-104    | p-101      | \$3,712.50 | \$1,222.00 | 33%    |
| 7                       | t-116 | 6/1/2023  | c-116     | d-104   | s-104    | p-101      | \$4,950.00 | \$1,065.00 | 22%    |
| 8                       | t-117 | 6/1/2023  | c-117     | d-103   | s-103    | p-101      | \$4,750.00 | \$810.00   | 17%    |
| 9                       | t-118 | 6/1/2023  | c-118     | d-103   | s-103    | p-101      | \$7,320.00 | \$933.00   | 13%    |
| 0                       | t-119 | 6/1/2023  | c-119     | d-104   | s-104    | p-101      | \$5,087.50 | \$655.00   | 13%    |
| 1                       | t-120 | 6/1/2023  | c-120     | d-104   | s-104    | p-101      | \$4,500.00 | \$722.00   | 16%    |
| 2                       | t-121 | 6/1/2023  | c-121     | d-104   | s-104    | p-105      | \$4,250.00 | \$901.00   | 21%    |
| 3                       | t-122 | 6/2/2023  | c-122     | d-104   | s-104    | p-102      | \$5,250.00 | \$1,349.00 | 26%    |
| 4                       | t-123 | 6/2/2023  | c-123     | d-102   | s-102    | p-102      | \$6,500.00 | \$1,288.00 | 20%    |
| 5                       | t-124 | 6/2/2023  | c-124     | d-102   | s-102    | p-102      | \$7,500.00 | \$1,664.00 | 22%    |
| 6                       | t-125 | 6/2/2023  | c-125     | d-102   | s-102    | p-101      | \$5,500.00 | \$1,320.00 | 24%    |
| 7                       | t-126 | 6/2/2023  | c-126     | d-102   | s-102    | p-101      | \$4,625.00 | \$1,001.00 | 22%    |
| 8                       | t-127 | 6/2/2023  | c-127     | d-102   | s-102    | p-101      | \$4,500.00 | \$960.00   | 21%    |
| 9                       | t-128 | 6/2/2023  | c-128     | d-102   | s-102    | p-105      | \$5,400.00 | \$540.00   | 10%    |
| 0                       | t-129 | 6/2/2023  | c-125     | d-102   | s-102    | p-101      | \$5,401.00 | \$500.00   | 9%     |

#### 4.Client\_Dim

|    | A                             | B           | C               | D | E |
|----|-------------------------------|-------------|-----------------|---|---|
| 1  |                               |             |                 |   |   |
| 2  | Client ▼                      | client_id ▼ | Contact ▼       |   |   |
| 3  | AMAZON.COM, INC.              | c-101       | Bill Smith      |   |   |
| 4  | AT&T INC.                     | c-119       | Bruno Cordova   |   |   |
| 5  | BANK OF AMERICA CORPORATION   | c-109       | Kylee Townsend  |   |   |
| 5  | CHEVRON CORPORATION           | c-118       | Denzel Flores   |   |   |
| 7  | CISCO SYSTEMS, INC.           | c-117       | Alia Thornton   |   |   |
| 8  | CITIGROUP INC.                | c-108       | Jordan Boone    |   |   |
| 9  | COMCAST CORPORATION           | c-123       | Bryce Carpenter |   |   |
| 10 | COSTCO WHOLESALE CORPORATION  | c-112       | Conor Wise      |   |   |
| 11 | DELL TECHNOLOGIES INC.        | c-124       | Jaidyn Andersen |   |   |
| 12 | EXXON MOBIL CORPORATION       | c-114       | Lucia Mckay     |   |   |
| 13 | FEDEX CORPORATION             | c-126       | Harry Lee       |   |   |
| 14 | GENERAL ELECTRIC COMPANY      | c-127       | Josh Johnson    |   |   |
| 15 | GENERAL MOTORS COMPANY        | c-121       | Bruce Rich      |   |   |
| 16 | INTEL CORPORATION             | c-120       | Jaylynn Napp    |   |   |
| 17 | JOHNSON & JOHNSON             | c-125       | Mark Walm       |   |   |
| 18 | JPMORGAN CHASE & CO.          | c-106       | Ivan Hiney      |   |   |
| 19 | LOCKHEED MARTIN CORPORATION   | c-128       | Mik Naam        |   |   |
| 20 | MCDONALD'S CORPORATION        | c-113       | Steven Michael  |   |   |
| 21 | MICROSOFT CORPORATION         | c-122       | Arturo Moore    |   |   |
| 22 | MORGAN STANLEY                | c-107       | Jonha Ma        |   |   |
| 23 | NETFLIX, INC.                 | c-103       | Harley Fritz    |   |   |
| 24 | TARGET CORPORATION            | c-111       | Brendan Wallace |   |   |
| 25 | TESLA, INC.                   | c-102       | Ken Singh       |   |   |
| 26 | THE GOLDMAN SACHS GROUP, INC. | c-105       | David Rasmussen |   |   |
| 27 | THE HOME DEPOT, INC.          | c-116       | Franklin Wrigt  |   |   |
| 28 | THE PROCTER & GAMBLE COMPANY  | c-104       | Nyla Novak      |   |   |
| 29 | VERIZON COMMUNICATIONS INC    | c-115       | Jose Roach      |   |   |
| 30 | WALMART INC.                  | c-110       | Nora Rollins    |   |   |
| 31 |                               |             |                 |   |   |
| 32 |                               |             |                 |   |   |
| 33 |                               |             |                 |   |   |

### 5.State\_Dim

|    | A          | B       |  |
|----|------------|---------|--|
| 1  |            |         |  |
| 2  | State      | state_i |  |
| 3  | California | s-104   |  |
| 4  | Florida    | s-103   |  |
| 5  | New York   | s-102   |  |
| 6  | Texas      | s-101   |  |
| 7  |            |         |  |
| 8  |            |         |  |
| 9  |            |         |  |
| 10 |            |         |  |
| 11 |            |         |  |

### 6.Department\_Dim

|   | A          | B       |  |
|---|------------|---------|--|
| 1 |            |         |  |
| 2 | Department | dept_id |  |
| 3 | Big Data   | d-104   |  |
| 4 | Cloud Tech | d-101   |  |
| 5 | Operations | d-103   |  |
| 6 | Strategy   | d-102   |  |
| 7 |            |         |  |
| 8 |            |         |  |

### 7.Payment\_Dim

|   |          |           |  |
|---|----------|-----------|--|
| 1 |          |           |  |
| 2 | Payment  | payment_i |  |
| 3 | Card     | p-105     |  |
| 4 | Check    | p-104     |  |
| 5 | Others   | p-103     |  |
| 6 | PayPal   | p-102     |  |
| 7 | Transfer | p-101     |  |
| 8 |          |           |  |
| 9 |          |           |  |